

六、练习题

4-1 图 4-43 所示结构中截面 A 转角（设顺时针为正）为（ ）。(哈尔滨工业大学 2009)

- A. $2F_P a^2/EI$ B. $-F_P a^2/EI$
C. $5F_P a^2/(4EI)$ D. $-5F_P a^2/(4EI)$

4-2 图 4-44 所示悬臂梁，B 点的竖向位移为_____。(合肥工业大学 2010)

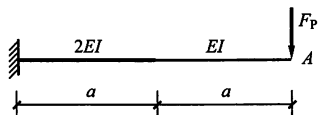


图 4-43

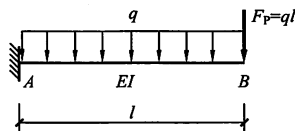


图 4-44

4-3 图 4-45 所示桁架，C 点的水平位移（ ）。(西南交通大学 2008)

- A. 向左 B. 向右



C. 等于零

D. 不定, 取决于 A_1/A_2 的值

4-4 图 4-46 所示刚架中, C、D 两点的相对线位移等于_____, 两点距离_____。(河北工业大学 2010)

4-5 图 4-47 所示桁架, 各杆拉伸刚度为 EA , DC 杆的转角为_____。(天津大学 2010)

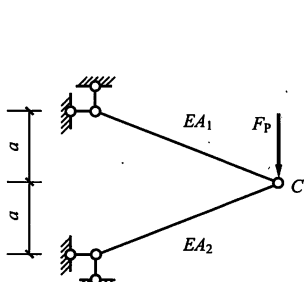


图 4-45

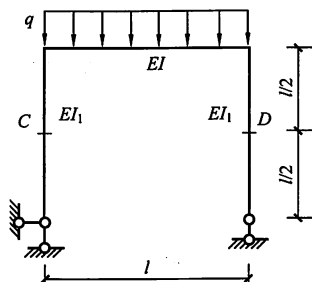


图 4-46

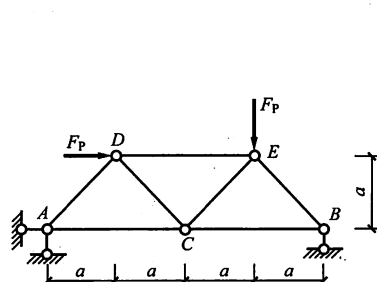


图 4-47

4-6 求图 4-48 所示结构 A、B 两点的相对水平位移, EI =常数。(中南大学 2012)

4-7 画出图 4-49 所示结构的弯矩图, 并求 C 点的水平位移。设各杆 EI 相同, 且忽略其轴向变形。(湖南大学 2012)

4-8 试求图 4-50 所示结构中截面 B 左右两侧的相对转角。 EI =常数。(福州大学 2010)

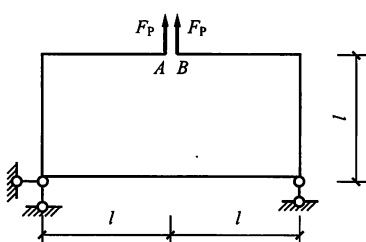


图 4-48

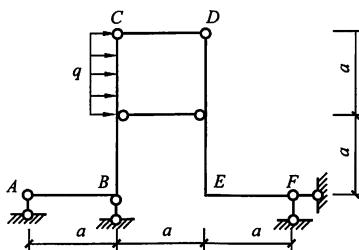


图 4-49

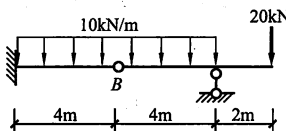


图 4-50

4-9 已知图 4-51 所示桁架各杆温度上升 $t^\circ\text{C}$, 材料的线膨胀系数为 α 。试求 K 点的竖向位移。(福州大学 2012)

4-10 计算图 4-52 所示结构中 A、B 两点相对水平线位移, EI =常数。(华南理工大学 2012)

4-11 计算图 4-53 所示荷载作用下的静定桁架。(1) 求各杆轴力; (2) 求结构 C 点的水平位移 Δ_{HC} 和竖向位移 Δ_{VC} 。已知: 各杆截面相同, 即 EA =常数。(长安大学 2008)

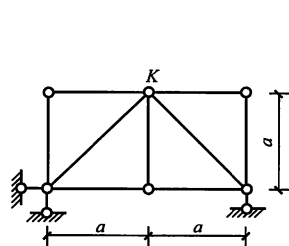


图 4-51

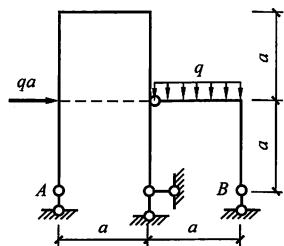


图 4-52

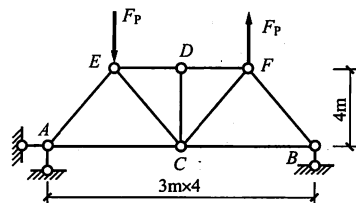


图 4-53



4-12 图 4-54 所示结构由于支座 A 沉降 1cm 引起截面 C 的转角 $\theta_c =$ _____。(重庆大学 2012)

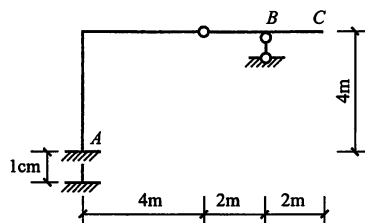


图 4-54

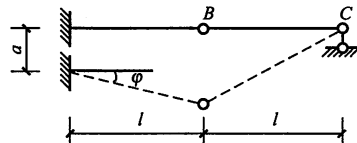


图 4-55

4-14 计算图 4-56 所示结构由于支座移动产生的铰 C 的角位移。(湖南大学 2008)

4-15 求图 4-57 所示结构 A 点竖向位移 Δ_{VA} , $EI =$ 常数。(西南交通大学 2007)

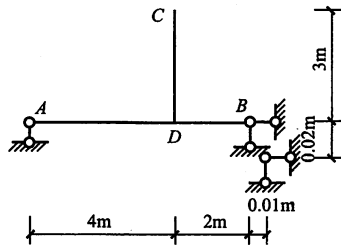


图 4-56

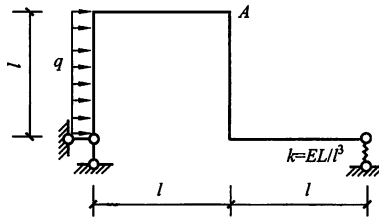


图 4-57

七、练习题答案

4-1 C。

4-2 $\frac{11ql^4}{24EI}$ ($\downarrow$)。

4-3 D。

4-4 $\frac{ql^4}{24EI}$; 增大。

4-5 $\frac{(2+\sqrt{2})F_P}{2EA}$ ($\rightarrow$)。

4-6 $\Delta_{AB} = \frac{3F_P l^3}{EI}$ ($\leftarrow \rightarrow$)。

4-7 弯矩图如图 4-58 所示, $\Delta_{HC} = \frac{67qa^4}{12EI}$ ($\rightarrow$)。

4-8 $\frac{320}{EI}$ ($\nearrow$)。

4-9 $-\alpha ta$ ($\uparrow$)。

4-10 $\frac{91qa^4}{24EI}$ ($\rightarrow \leftarrow$)。

4-11 各杆轴力如图 4-59 所示。 $\Delta_{HC} = \frac{9F_P}{4EA}$ ($\rightarrow$); $\Delta_{VC} = 0$ (由反对称荷载下位移的



The diagram shows a truss structure with a horizontal bottom chord and a top chord. The bottom chord has four joints: a fixed support on the left, a roller support in the middle, and a roller support on the right. The top chord has three joints. A downward force F_P is applied at the left top joint, and an upward force F_P is applied at the right top joint. The horizontal distance between the left and middle bottom joints is $3/8 F_P$, and between the middle and right bottom joints is $3/8 F_P$. The vertical height of the truss is $5/8 F_P$. The internal forces in the members are labeled: the leftmost diagonal member has a force of $5/8 F_P$ (compression), the diagonal member connecting the left top joint to the middle bottom joint has a force of $5/8 F_P$ (tension), the diagonal member connecting the middle bottom joint to the right top joint has a force of $5/8 F_P$ (tension), and the rightmost diagonal member has a force of $5/8 F_P$ (compression). The horizontal members (top and bottom) have a force of 0 . The vertical member connecting the middle bottom joint to the right top joint has a force of 0 .

4-12 0.005rad (↙)。
 4-13 0.003rad (↘)。
 4-14 $\varphi_C=0.003\text{ 3rad}$ (↘)。
 4-15 $\Delta_{VA}=\frac{3ql^4}{8EI}$ (↓)。